1.3 Using Midpoint and Distance Formulas

Name _____ Period ____ Date _____

Geometry 1.3 · Student sheet

I can…

- use a midpoint or segment bisector to find lengths.
- use the Midpoint Formula to find a midpoint or an endpoint.
- use the Distance Formula to find the length of a segment.

Vocabulary · fill in the missing word

Midpoint — the point that divides a segment into two _____ segments

Segment bisector — a point, ray, line, segment, or plane that _____ a segment at its midpoint

Part 1 · Discovery: halfway and how far

Plot A(1, 2) and B(7, 10) and draw $\overline{AB}$. Find the middle with a ruler. Then draw the right triangle under $\overline{AB}$.

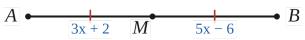
Middle point	Average of the x's	Average of the y's	Horizontal leg	Vertical leg	AB

What do you notice?

Part 2 · Notes and examples (fill in each note, then work the example)

Example 1

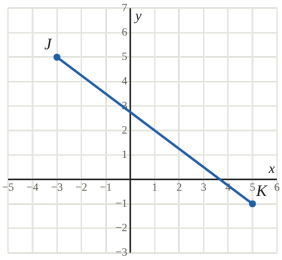
M is the midpoint of $\overline{AB}$. AM = 3x + 2 and MB = 5x − 6. Find x and AB.



Midpoint Formula: M = (_____, _____)

Example 2

Find the midpoint of $\overline{JK}$ with J(−3, 5) and K(5, −1).



Finding an endpoint: the midpoint is _____ between the endpoints, so add the same change again.

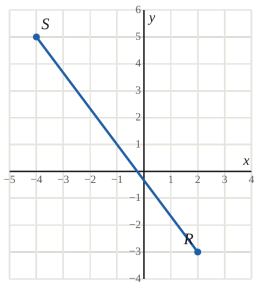
Example 3

M(2, 1) is the midpoint of $\overline{AB}$, and A is at (−1, 4). Find B.

Distance Formula: AB = _____

Example 4

Find the distance between R(2, −3) and S(−4, 5).



Example 5

On a map grid in miles, your home is at (−2, 3) and school is at (4, −5). A library is exactly halfway between them. Where is the library, and how far is it from home to school?

Part 3 · Partner practice

Partner 1

Find the midpoint of the segment with endpoints (4, −6) and (−2, 8).

Partner 2

Find the distance between (1, −2) and (6, 10).